Data Flow Diagram

Metadata

- **Artifact ID**: data_flow_diagram_Analyst_v1.0
- **Version**: 1.0
- **Created By**: Analyst
- **Timestamp**: 2026-03-19T06:38:11.725371+00:00

Diagrams

Diagram 1: Diagram Base64

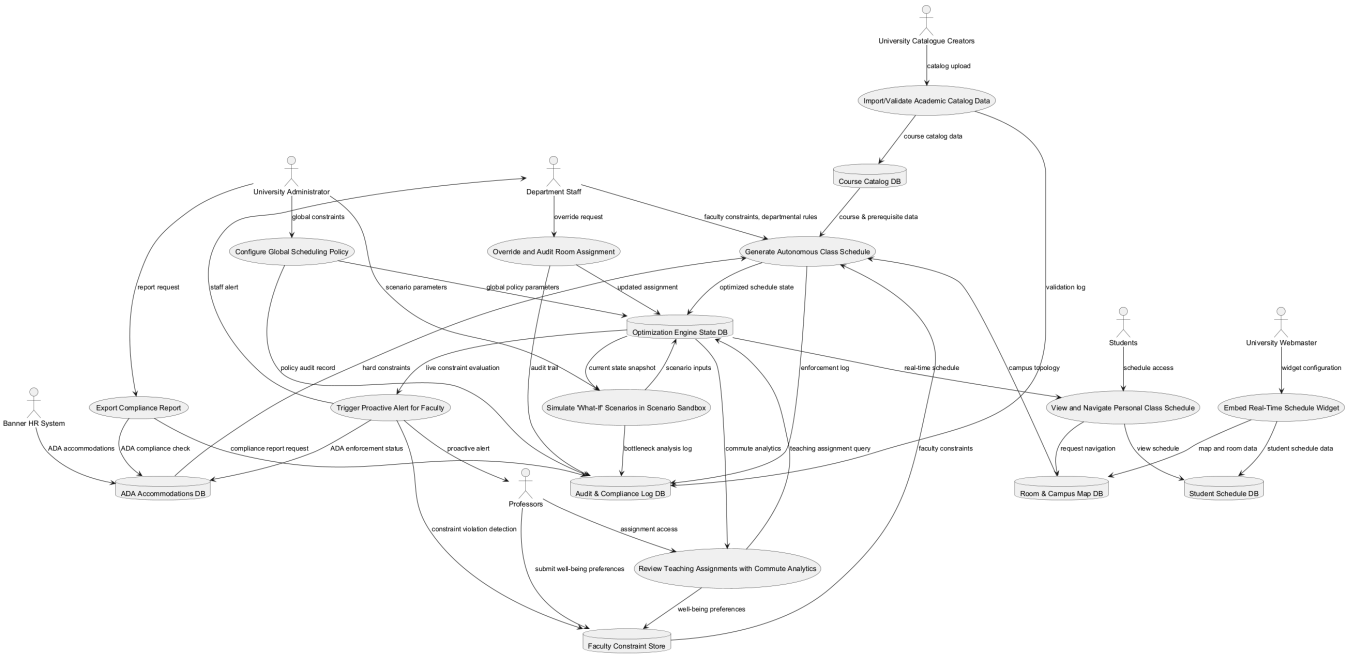

Diagram Content

External entities: Department Staff, University Administrator, University Catalogue Creators, University Webmaster, Students, Professors, Banner HR System. Processes: Generate Autonomous Class Schedule, Configure Global Scheduling Policy, Import/Validate Academic Catalog Data, Embed Real-Time Schedule Widget, View and Navigate Personal Class Schedule, Review Teaching Assignments with Commute Analytics, Simulate 'What-If' Scenarios in Scenario Sandbox, Override and Audit Room Assignment, Export Compliance Report, Trigger Proactive Alert for Faculty. Data stores: Faculty Constraint Store, Course Catalog DB, Room & Campus Map DB, Optimization Engine State DB, Audit & Compliance Log DB, ADA Accommodations DB, Student Schedule DB. Key data flows include: Department Staff submitting constraints and reviewing schedules → Generate Autonomous Class Schedule process; University Administrator configuring policies → Configure Global Scheduling Policy process; Banner HR feeding ADA accommodations → ADA Accommodations DB → Optimization Engine; Catalogue Creators importing course data → Course Catalog DB → validation and scheduling logic; Students and Professors accessing personalized schedules → Student Schedule DB and Optimization Engine State DB; Webmaster embedding widgets → Room & Campus Map DB and Student Schedule DB; Scenario Sandbox consuming enrollment forecasts and faculty availability → generating alerts and alternatives; all constraint changes and enforcement decisions → Audit & Compliance Log DB.

Uml Fmt Content

```
@startuml
actor "Department Staff" as E1
actor "University Administrator" as E2
actor "University Catalogue Creators" as E3
actor "University Webmaster" as E4
actor "Students" as E5
actor "Professors" as E6
actor "Banner HR System" as E7

usecase "Generate Autonomous Class Schedule" as P1
usecase "Configure Global Scheduling Policy" as P2
usecase "Import/Validate Academic Catalog Data" as P3
usecase "Embed Real-Time Schedule Widget" as P4
usecase "View and Navigate Personal Class Schedule" as P5
usecase "Review Teaching Assignments with Commute Analytics" as P6
usecase "Simulate 'What-If' Scenarios in Scenario Sandbox" as P7
usecase "Override and Audit Room Assignment" as P8
usecase "Export Compliance Report" as P9
usecase "Trigger Proactive Alert for Faculty" as P10

database "Faculty Constraint Store" as D1
database "Course Catalog DB" as D2
database "Room & Campus Map DB" as D3
database "Optimization Engine State DB" as D4
database "Audit & Compliance Log DB" as D5
database "ADA Accommodations DB" as D6
database "Student Schedule DB" as D7

E1 --> P1 : "faculty constraints, departmental rules"
E1 --> P8 : "override request"
P1 --> D4 : "optimized schedule state"
P1 --> D5 : "enforcement log"
P2 --> D4 : "global policy parameters"
P2 --> D5 : "policy audit record"
P3 --> D2 : "course catalog data"
P3 --> D5 : "validation log"
P4 --> D3 : "map and room data"
P4 --> D7 : "student schedule data"
P5 --> D7 : "view schedule"
P5 --> D3 : "request navigation"
P6 --> D4 : "teaching assignment query"
P6 --> D1 : "well-being preferences"
P7 --> D4 : "scenario inputs"
P7 --> D5 : "bottleneck analysis log"
P8 --> D4 : "updated assignment"
P8 --> D5 : "audit trail"
P9 --> D5 : "compliance report request"
P9 --> D6 : "ADA compliance check"
P10 --> D1 : "constraint violation detection"
P10 --> D6 : "ADA enforcement status"
P10 --> E6 : "proactive alert"
P10 --> E1 : "staff alert"
E2 --> P2 : "global constraints"
E2 --> P7 : "scenario parameters"
E2 --> P9 : "report request"
```

```
E3 --> P3 : "catalog upload"
E4 --> P4 : "widget configuration"
E5 --> P5 : "schedule access"
E6 --> P6 : "assignment access"
E6 --> D1 : "submit well-being preferences"
E7 --> D6 : "ADA accommodations"
D6 --> P1 : "hard constraints"
D1 --> P1 : "faculty constraints"
D2 --> P1 : "course & prerequisite data"
D3 --> P1 : "campus topology"
D4 --> P5 : "real-time schedule"
D4 --> P6 : "commute analytics"
D4 --> P7 : "current state snapshot"
D4 --> P10 : "live constraint evaluation"
@enduml
```

Summary

The DFD models data flows among six stakeholder roles, core scheduling processes, and seven critical data stores—emphasizing autonomous optimization, real-time synchronization, proactive bottleneck resolution, and strict compliance enforcement across faculty constraints, ADA accommodations, and FERPA-protected data.